

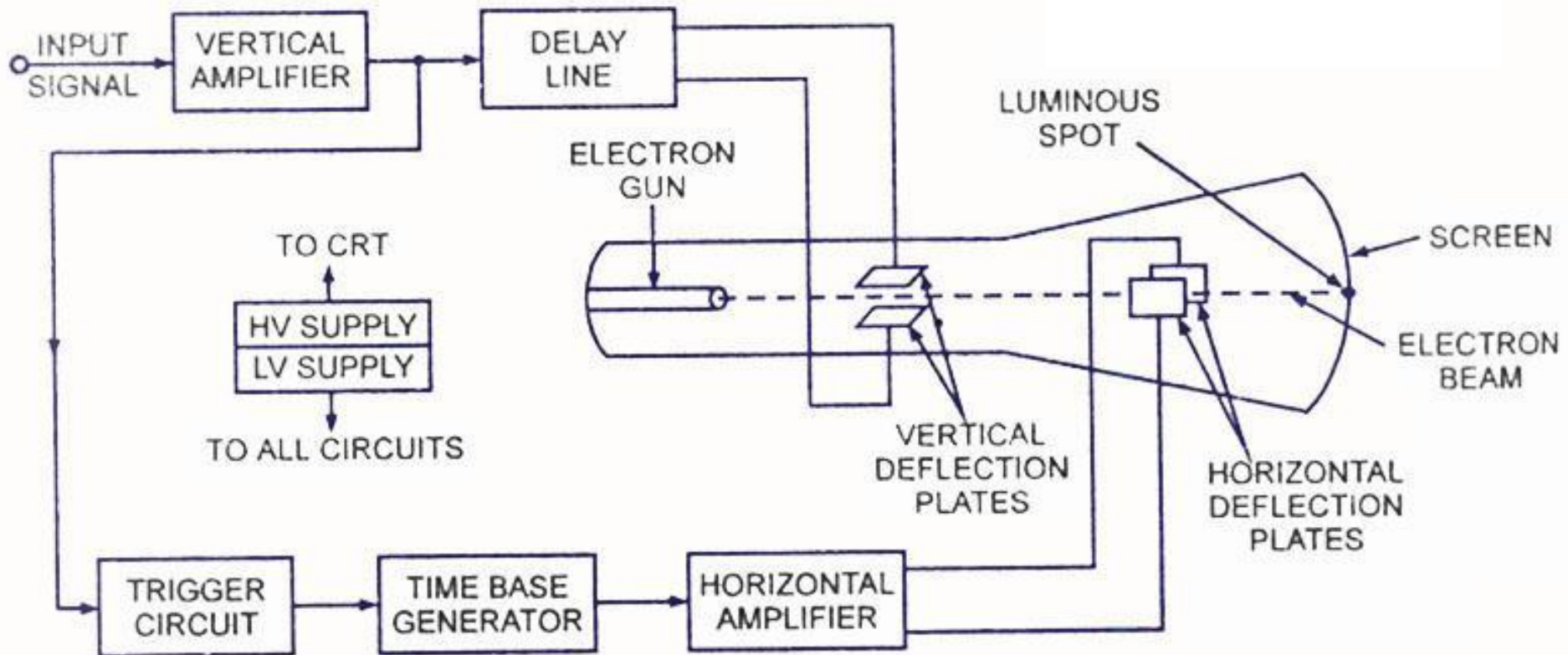
Chapter-4-

Cathode Ray Oscilloscope

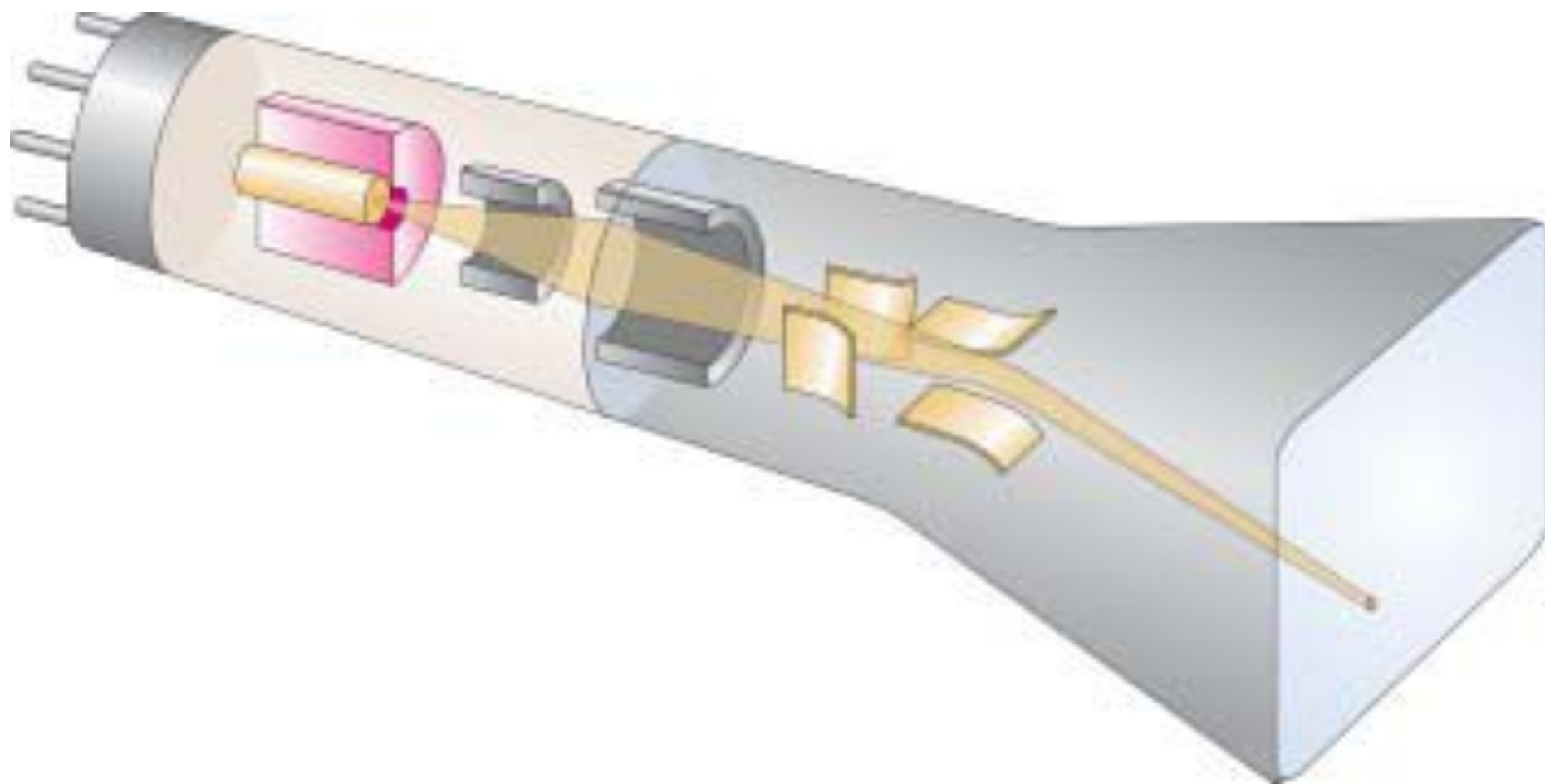
Introduction to Cathode Ray Oscilloscope

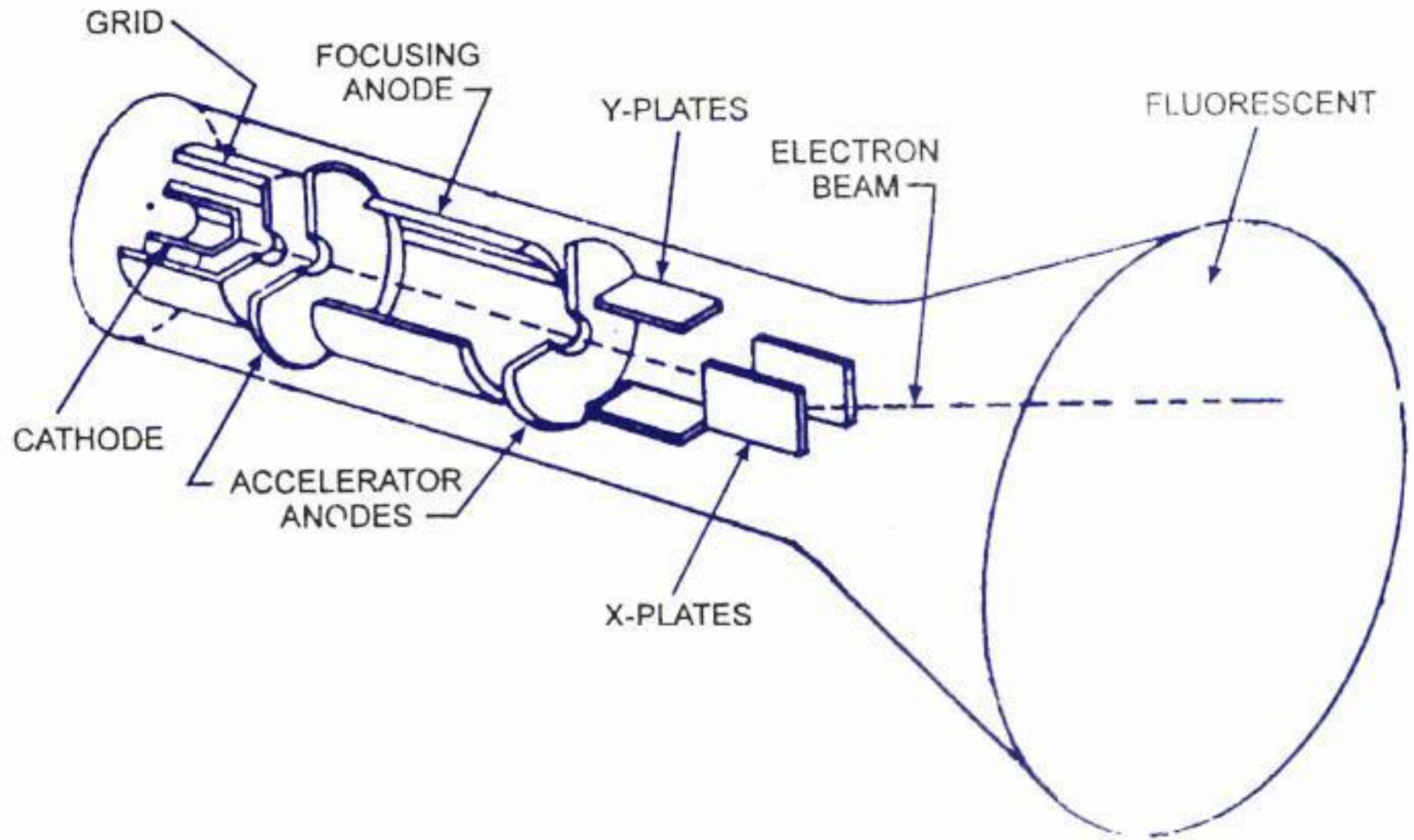
The *cathode ray oscilloscope (CRO)* is an extremely useful and versatile laboratory instrument used for studying wave shapes of alternating currents and voltages as well as for measurement of voltage, current, power and frequency, in fact, almost any quantity that involves amplitude and waveform. It allows the user to see the amplitude of electrical signals as a function of time on the screen

Block Diagram of a CRO



Block Diagram of a General Purpose CRO





Cathode Ray Tube

Cathode ray tube essentially consists of an electron gun for producing a stream of electrons, focusing and accelerating anodes for producing a narrow and sharply focused electron beam, horizontal and vertical deflection plates for controlling the beam path and an evacuated glass envelope with phosphorescent screen giving bright spot when struck by a high velocity electron beam.

The deflection system

The beam may be moved in the X- and Y-directions by using the time base system.

The time base circuit applies a saw-tooth waveform to the X-plates, as shown in Figure 2.

The beam is moved from the left-hand side of the screen to the right during the time that the voltage rises to a maximum, and then is returned rapidly to the left as the voltage returns to zero. This fly-back time should be as short as possible.

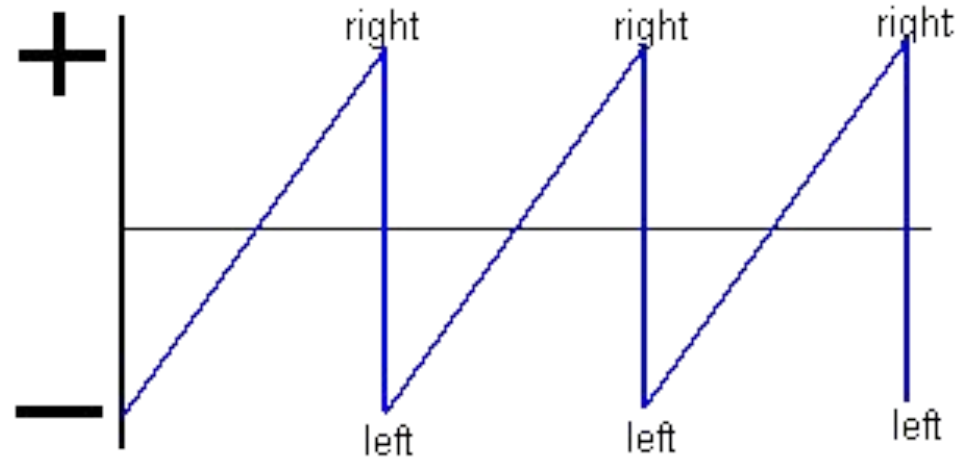
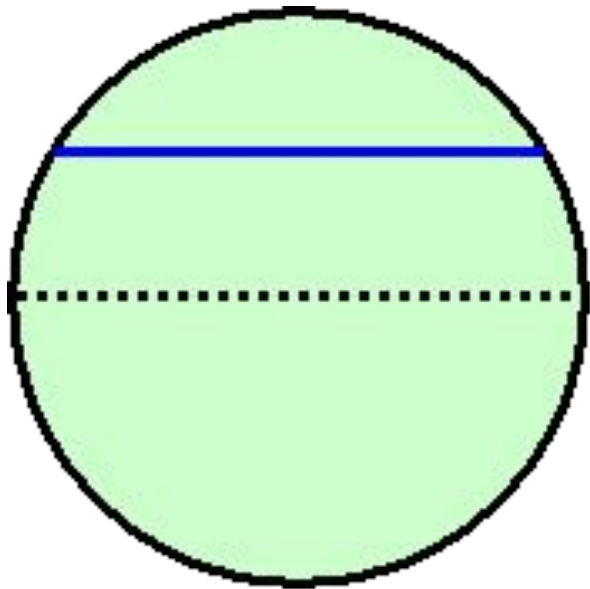
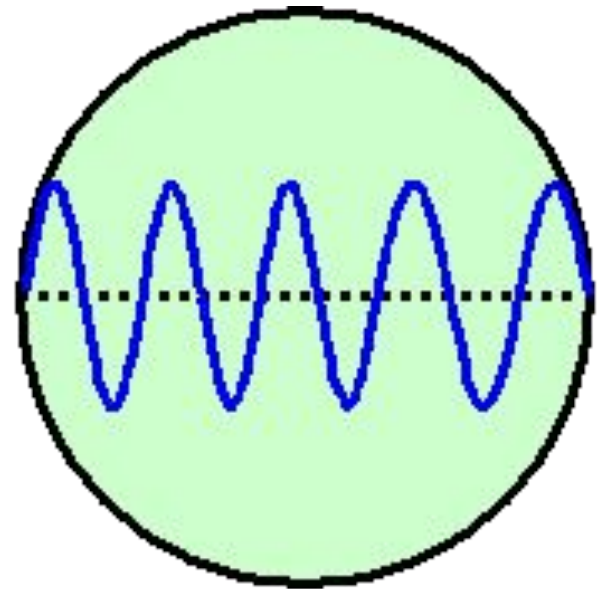


Figure 2

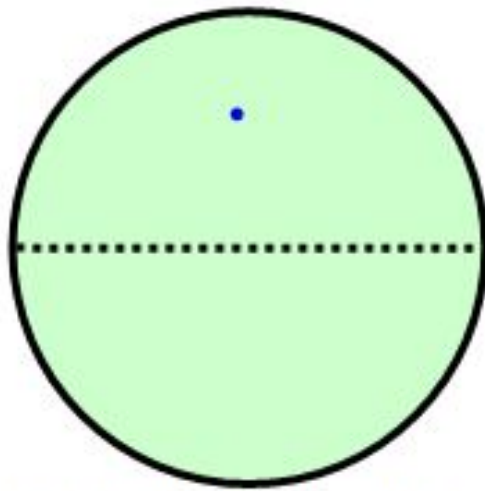
If a voltage is now applied to the Y-inputs, the variation of this voltage with time may be displayed on the screen.



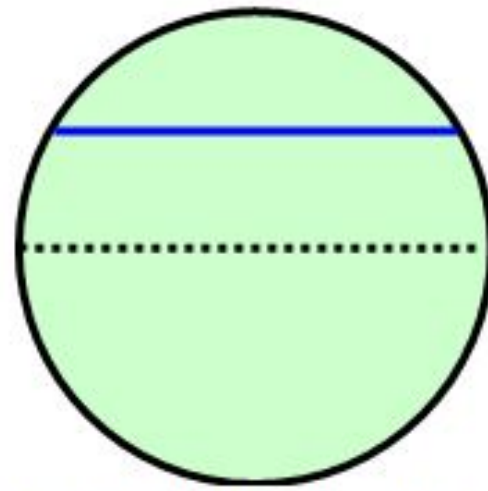
d.c. input



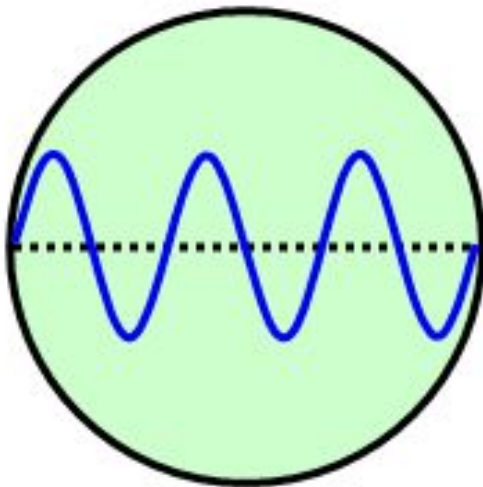
a.c. input



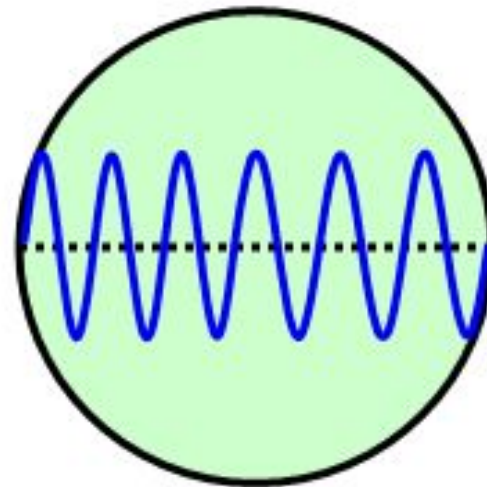
d.c. input with the time base off



d.c. input with the time base on



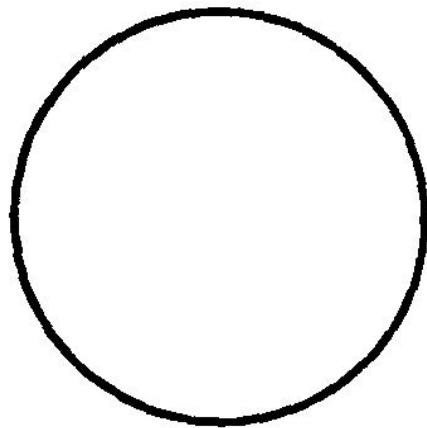
a.c. input with a fast time base



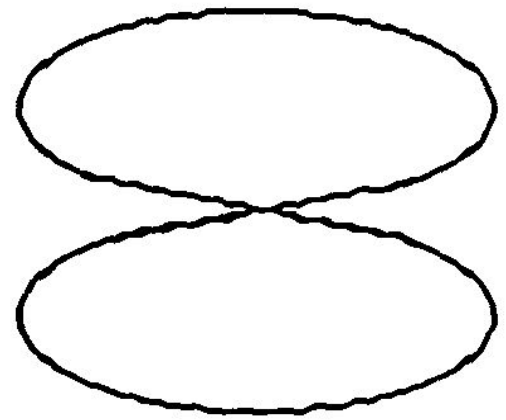
a.c. input with a slow time base

Lissajous Figures:

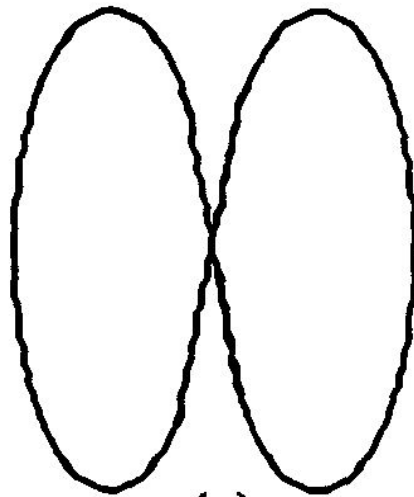
When sine-wave signals of different frequencies are input to the horizontal and vertical amplifiers a stationary pattern is formed on the CRT when the ratio of the two frequencies is an integral fraction such as $1/2$, $1/3$, $1/4$, $1/5$, etc. These stationary patterns are known as *Lissajous figures* and can be used for comparison measurement of frequencies.



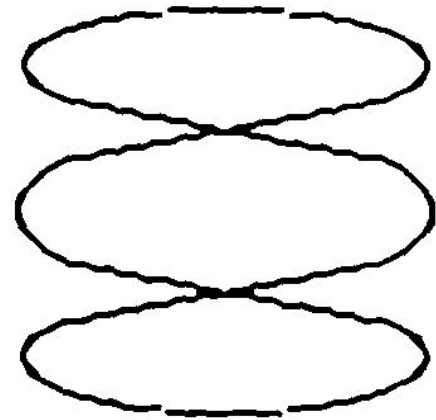
(a)



(b)

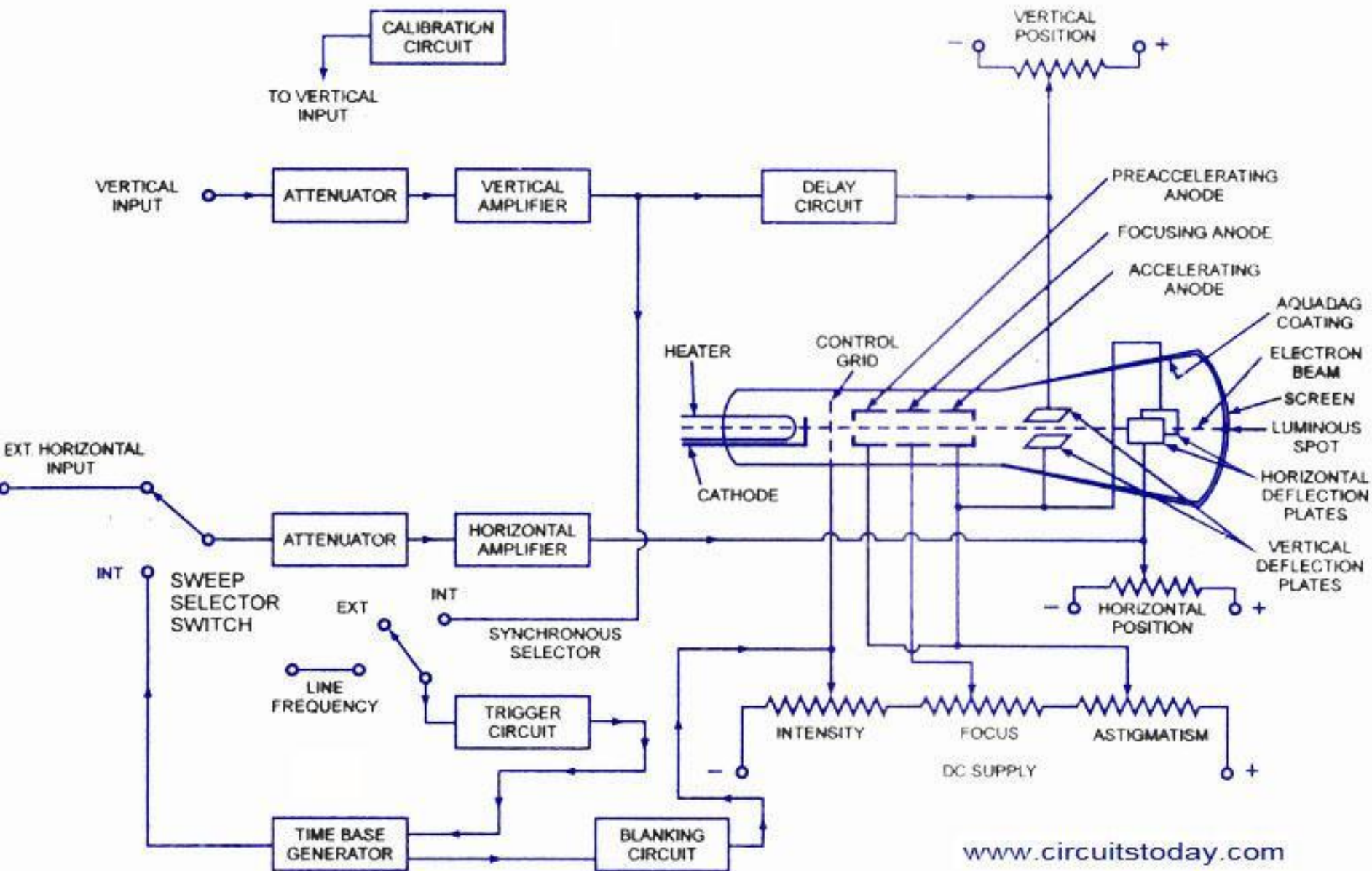


(c)



(d)

Figure 5. Lissajous figures for horizontal-to-vertical frequency ratios of: (a) 1:1, (b) 2:1, (c) 1:2, and (d) 3:1.



Detailed Block Diagram of a CRO